Calculating CPTG Galaxy Rotation Curves

DDO-154, NGC-3198, and NGC-7814

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Abstract

This tutorial presents a step-by-step calculation of the Curvature Polarization Transport Gravity (CPTG) galaxy-limit equation using three worked rotation-curve examples: DDO-154, NGC-3198, and NGC-7814. Starting from SPARC-style rotation-curve inputs, the calculation converts the observed radius and velocity columns into physical acceleration fields, builds the CPTG baryonic source field, infers the object-dependent structural acceleration scale $a_\star$, derives the structural transport scale, computes the transport remnant, solves the implicit nonlinear acceleration equation, and recovers the model rotation curve.

The three galaxies are treated separately. DDO-154 is a gas-rich dwarf system and a classic low-acceleration test case for dark matter and modified-gravity theories. NGC-3198 is a large spiral galaxy with an extended, nearly flat H I rotation curve and has long served as a standard reference case for disk-galaxy mass discrepancies. NGC-7814 is an edge-on, bulge-dominated spiral whose rotation-curve decomposition includes a large stellar bulge contribution. Together, the three examples show how the same locked CPTG galaxy calculation is carried through a low-mass dwarf, an extended disk, and a centrally concentrated bulge system. A fixed-a_0, unit-source MOND/RAR-form calculation is retained as a quantitative contrast so that the different source mapping, acceleration law, predicted curve, and residual diagnostics can be followed from the same SPARC component inputs. The purpose of the paper is instructional: to make the CPTG rotation-curve calculation reproducible step by step.

1 Introduction

Galaxy rotation curves provide a direct way to compare an observed radial acceleration profile with the gravitational response calculated from the visible matter in a galaxy. In Curvature Polarization Transport Gravity (CPTG), that calculation begins with the resolved baryonic source rather than with an added dark-matter halo. The gas, stellar disk, and stellar bulge contributions are mapped through the locked CPTG galaxy-source coefficients, and the resulting source is carried through the same nonlinear galaxy-response law for every system. The quantity that changes from one galaxy to another is the structural acceleration scale $a_\star$, which characterizes the gravitating system and sets the strength of its CPTG response.

The practical calculation therefore has a clear sequence. The observed source decomposition is converted into a CPTG baryonic source field; one object-dependent $a_\star$ is inferred for the galaxy; the associated transport scale follows from that value; and the nonlinear field equation is solved across the measured radii. The solved acceleration is then converted back into a circular-velocity curve. The post-solution Structural Mode is read only after this dynamical solution has been obtained, so the structural classification does not participate in fitting the rotation curve.

MOND is included throughout the tutorial as a familiar comparison resource. Its use in galaxy rotation-curve work dates to the early MOND literature of the 1980s [7], and many readers already know the basic idea of relating the baryonic acceleration to an enhanced gravitational response through a universal acceleration scale. That familiarity makes MOND useful for showing where the two calculation architectures differ. Both calculations start from the same SPARC component data, but CPTG applies its locked source mapping and an object-dependent structural scale, whereas the comparison used here applies an explicitly defined unit-source baryonic field with fixed universal a_0. This fixed-a_0, unit-source MOND/RAR-form calculation is the comparison convention adopted for this tutorial; it is not presented as a unique MOND fitting prescription. The comparison curves and residual statistics are therefore included to make the numerical differences transparent, not to change the instructional purpose of the paper.

The three worked examples were chosen because they expose different parts of the same calculation. DDO-154 shows the weak-source, gas-rich dwarf regime; NGC-3198 shows an extended disk with a long nearly flat rotation curve; and NGC-7814 shows how the same CPTG procedure handles a strong stellar bulge. The sections that follow keep the derivation explicit enough that each stage can be reproduced directly from the listed rotation-curve inputs.

2 Core Galaxy-Limit Equation

The reduced CPTG [9] galaxy equation is an implicit nonlinear relation for the total radial acceleration field:

$$g(r) = g_{\mathrm{bar,CPTG}}(r) + \frac{\sqrt{a_\star g_{\mathrm{bar,CPTG}}(r)} + g_{\mathrm{geom}}(r)}{\left(1 + (g(r)/a_\star)^{123/50}\right)^{11/50}}. \tag{1}$$

The geometric transport remnant is

$$g_{\mathrm{geom}}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \tag{2}$$

The galaxy-limit transport coefficient is fixed as

$$\eta = \frac{1}{2}. \tag{3}$$

The structural transport scale is derived from the inferred structural acceleration scale:

$$R_c = \frac{48\pi c^2}{a_\star}. \tag{4}$$

The transport scale is therefore a derived quantity once $a_\star$ has been inferred.

3 The Structural Acceleration Scale $a_\star$

The structural acceleration scale $a_\star$ is the object-dependent acceleration scale associated with the gravitating system in the CPTG galaxy reduction. It is a system-level structural quantity: for a given galaxy calculation one value of $a_\star$ is inferred from the resolved source and rotation curve, and that value is then used at every measured radius. It is therefore distinct from the radial fields $g_{\mathrm{bar,CPTG}}(r)$, $g_{\mathrm{obs}}(r)$, and $g_{\mathrm{CPTG}}(r)$. In the general CPTG framework $a_\star \geq 0$; the normalized galaxy equations used below are written on the positive branch $a_\star > 0$. The admissible state $a_\star = 0$ is the continuous baryonic endpoint, for which $R_c \to \infty$ and $g(r) \to g_{\mathrm{bar,CPTG}}(r)$. The three worked galaxies considered here have positive inferred values.

The distinction between $a_\star$ and the locked galaxy law is fundamental. The gas, stellar-disk, and stellar-bulge source coefficients 27/50, 23/50, and 16/25, the transport coefficient $\eta = 1/2$, the transport relation, and the nonlinear exponents are fixed theory quantities and do not vary from galaxy to galaxy. The object-dependent quantity is $a_\star$.

Operationally, a trial $a_\star$ changes two linked parts of the CPTG calculation. It sets the amplitude of the square-root polarization contribution,

$$g_{\mathrm{pol}}(r) = \sqrt{a_\star g_{\mathrm{bar,CPTG}}(r)}, \tag{5}$$

and at the same time fixes the structural transport radius through Eq. (4).

The implicit CPTG field equation is solved across the full measured rotation curve for each trial value, and the retained $a_\star$ is the value that minimizes the error-weighted velocity residual statistic defined below. Once $a_\star$ is retained, R_c is derived rather than fitted independently.

This differs from the universal-acceleration-scale construction used in the MOND calculation shown in this tutorial. The MOND curve retains

$$a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2} \tag{6}$$

for every galaxy, whereas CPTG assigns the gravitating system its own structural acceleration scale $a_\star$ while preserving the same locked galaxy reduction. The fact that $a_\star$ and a_0 have the same physical dimensions does not give them the same theoretical status: a_0 is universal in the MOND calculation used here; $a_\star$ is object-dependent in CPTG.

4 Input Data and Unit Conversion

Each rotation-curve file is read as a SPARC-style table following the database conventions of Lelli, McGaugh, and Schombert [5]. The required columns are radius, observed velocity, velocity uncertainty, gas velocity contribution, stellar disk velocity contribution, and stellar bulge velocity contribution. In symbolic form, the required columns are

$$r_{\text{kpc}}, \qquad V_{\text{obs}}, \qquad \sigma_V, \qquad V_{\text{gas}}, \qquad V_{\text{disk}}, \qquad V_{\text{bulge}}. \tag{7}$$

The radius is converted to SI units by

$$r = r_{\text{kpc}} \left(3.085677581491367 \times 10^{19}\right) \text{ m}. \tag{8}$$

Velocities are converted by

$$V = V_{\text{km/s}} \left(10^3\right) \text{ m s}^{-1}. \tag{9}$$

The observed centripetal acceleration is then

$$g_{\text{obs}}(r) = \frac{V_{\text{obs}}^2(r)}{r}. \tag{10}$$

The observed acceleration is not inserted into the pointwise CPTG field equation. The observed rotation curve enters through the global inference of $a_\star$ and through the residual checks after the theoretical field has been calculated.

5 Constructing the CPTG Baryonic Source

CPTG uses a locked galaxy-source mapping. The gas, stellar-disk, and stellar-bulge coefficients are 27/50, 23/50, and 16/25, respectively, and these coefficients do not vary from galaxy to galaxy. The source is built from the component velocity fields using those locked coefficients. When a SPARC component velocity carries a sign, the signed contribution is preserved through the signed-square convention

$$Q(V) = V|V|. \tag{11}$$

For positive component velocities this reduces to the ordinary squared term:

$$Q(V) = V^2. \tag{12}$$

The signed weighted source sum is

$$\widetilde{V}^2_{\mathrm{bar,CPTG}}(r) = \frac{27}{50} Q\left(V_{\mathrm{gas}}(r)\right) + \frac{23}{50} Q\left(V_{\mathrm{disk}}(r)\right) + \frac{16}{25} Q\left(V_{\mathrm{bulge}}(r)\right). \tag{13}$$

The scalar radial source supplied to the galaxy equation is nonnegative:

$$V^2_{\mathrm{bar,CPTG}}(r) = \max\left(\widetilde{V}^2_{\mathrm{bar,CPTG}}(r), 0\right). \tag{14}$$

For the three worked examples the net weighted source remains positive at the displayed radii. The CPTG baryonic acceleration field is therefore

$$g_{\mathrm{bar,CPTG}}(r) = \frac{V^2_{\mathrm{bar,CPTG}}(r)}{r}. \tag{15}$$

For calculation contrast only, the MOND source uses unit component coefficients:

$$V^2_{\mathrm{bar,MOND}}(r) = Q\left(V_{\mathrm{gas}}(r)\right) + Q\left(V_{\mathrm{disk}}(r)\right) + Q\left(V_{\mathrm{bulge}}(r)\right). \tag{16}$$

The corresponding baryonic acceleration is

$$g_{\mathrm{bar,MOND}}(r) = \frac{V^2_{\mathrm{bar,MOND}}(r)}{r}. \tag{17}$$

The fixed-a_0, unit-source comparison shown in the figures applies the RAR-form acceleration response

$$g_{\mathrm{MOND}}(r) = \frac{g_{\mathrm{bar,MOND}}(r)}{1 - \exp\left[-\sqrt{g_{\mathrm{bar,MOND}}(r)/a_0}\right]}, \qquad a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}, \tag{18}$$

using the functional form associated with the empirical radial-acceleration relation of McGaugh, Lelli, and Schombert [6]. The source normalization in Eq. (16) is the explicit unit-source convention adopted for this tutorial; it is not an assertion that all MOND analyses use the same stellar normalization or fitting procedure. This calculation shows how the same SPARC component inputs are mapped through a different source prescription and acceleration law. The comparison curve and residual statistics are retained alongside the CPTG calculation so that the complete numerical contrast can be followed.

The separation between the two source fields matters. In CPTG, the component velocities are used once to build the CPTG baryonic source. After that, the nonlinear CPTG equation is solved for the total field directly.

6 Solving the Galaxy Equation

6.1 Step 1: Determine the Structural Acceleration Scale

For a trial value of the structural acceleration scale, the structural transport scale is first computed from Eq. (4).

The nonlinear galaxy equation is then solved for the total acceleration field. The predicted circular velocity follows from

$$V_{\rm CPTG}(r) = \sqrt{g(r)r}. \tag{19}$$

The retained value of $a_\star$ is the value that minimizes the rotation-curve residual statistic:

$$\chi^2_{\rm CPTG} = \sum_i \left(\frac{V_{{\rm obs},i} - V_{{\rm CPTG},i}}{\sigma_{V,i}} \right)^2 . \tag{20}$$

Because $a_\star$ is the one inferred quantity in the CPTG rotation-curve calculation, the reported CPTG reduced statistic uses

$$\nu_{\rm CPTG} = N_{\rm data} - 1, \qquad \chi^2_{\nu,{\rm CPTG}} = \frac{\chi^2_{\rm CPTG}}{N_{\rm data} - 1}. \tag{21}$$

For either calculated velocity curve $V_{\rm model}$, the residual summaries used below are

$$\mathrm{RMS} = \sqrt{\frac{1}{N_{\rm data}} \sum_i \left(V_{{\rm obs},i} - V_{{\rm model},i}\right)^2}, \qquad \mathrm{MAE} = \frac{1}{N_{\rm data}} \sum_i \left|V_{{\rm obs},i} - V_{{\rm model},i}\right| . \tag{22}$$

The χ^2 statistics use the quoted pointwise SPARC velocity uncertainties; RMS and MAE are unweighted velocity-residual summaries. These diagnostics are not a full likelihood with covariance, distance, inclination, or other observational systematics marginalized as additional nuisance parameters.

To state the model–data agreement directly as a fraction of the observed circular speed, define the absolute fractional velocity residual

$$\epsilon_{V,i} = \frac{|V_{{\rm model},i} - V_{{\rm obs},i}|}{V_{{\rm obs},i}}, \qquad \epsilon_{V,{\rm med}} = 100\ \mathrm{median}_i(\epsilon_{V,i}) , \tag{23}$$

where $\epsilon_{V,{\rm med}}$ is reported in percent. The numerical summaries also report how many measured radii satisfy $\epsilon_{V,i} \le 0.10$. These are unweighted descriptive agreement measures and do not replace the uncertainty-weighted χ^2 statistic.

6.2 Step 2: Solve the Implicit Field Equation

At each radius, the acceleration field appears on both sides of the equation. It is therefore solved iteratively. In the tutorial sections below, the baryonic term is kept explicitly as the CPTG source field rather than shortened to a generic symbol.

A useful way to write the iteration is

$$g_{n+1}(r) = g_{\mathrm{bar,CPTG}}(r) + \left(\sqrt{a_\star g_{\mathrm{bar,CPTG}}(r)} + g_{\mathrm{geom}}(r)\right)\Sigma_n(r). \tag{24}$$

The nonlinear suppression factor at iteration step number n is

$$\Sigma_n(r) = \left(1 + \left(\frac{g_n(r)}{a_\star}\right)^{123/50}\right)^{-11/50}. \tag{25}$$

The dominant curvature-polarization enhancement is $g_{\mathrm{pol}}(r)$ as defined in Eq. (5).

The total nonlinear enhancement added to the CPTG baryonic source is

$$\Delta g(r) = (g_{\mathrm{pol}}(r) + g_{\mathrm{geom}}(r))\,\Sigma(r). \tag{26}$$

The final CPTG acceleration field is

$$g_{\mathrm{CPTG}}(r) = g_{\mathrm{bar,CPTG}}(r) + \Delta g(r). \tag{27}$$

6.3 Numerical Implementation

For each trial $a_\star > 0$, the registered numerical implementation initializes the fixed-point solve with $g_0(r) = g_{\mathrm{bar,CPTG}}(r)$. It performs at most 300 iterations and stops when

$$\max_r \frac{|g_{n+1}(r) - g_n(r)|}{\max(|g_{n+1}(r)|\,, 10^{-30})} < 10^{-13}. \tag{28}$$

The structural scale is searched in $x = \log_{10}(a_\star/\mathrm{m\,s^{-2}})$. The registered search begins on $-12 \le x \le -9$ with bounded scalar minimization at 10^{-8} log-space tolerance, then evaluates a 161-point refinement over a ± 0.02 dex neighborhood of the provisional minimum. If the refined minimum is clipped within 0.005 dex of a search boundary and extending that boundary by one decade lowers χ^2 at six-decimal precision, the search is extended outward and repeated, with at most 12 one-decade extensions on either side. Because a logarithmic search cannot represent $a_\star = 0$, that admissible state is the separate continuous baryonic endpoint described above rather than an interior trial value.

Additional digits are retained in the worked values to identify and reproduce the deterministic numerical solution. They do not represent observational precision in $a_\star$, R_c, or the Structural Mode estimator.

7 Reading the Structural Mode After the Solve

The Structural Mode is a post-solution measure of the radial organization of the solved CPTG response. It is evaluated only after the acceleration field has been obtained and does not enter the rotation-curve fit. The continuum functional and its registered sampled-data implementation are distinguished explicitly here, following the parent CPTG framework [9].

The reduced galaxy diagnostic uses the fixed structural-gradient scale

$$\kappa_{\text{struct}} = 1 \text{ kpc}. \tag{29}$$

This fixed scale provides a consistent radial normalization for the gradient term across galaxy calculations. In the continuum definition, the solved field produces

$$C(r) = g^2(r) + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr}\right)^2, \tag{30}$$

with outer resolved size

$$R = \max(r), \tag{31}$$

and curvature-structure weight

$$w(r) = \left|\frac{dC}{dr}\right|. \tag{32}$$

The continuum characteristic length and Structural Mode are therefore

$$\lambda = \frac{\int_{R/5}^{R} r\, w(r)\, dr}{\int_{R/5}^{R} w(r)\, dr}, \qquad N = \frac{R}{\lambda}. \tag{33}$$

A SPARC rotation curve is a sampled radial profile rather than a continuous field. The registered Workbench estimator first evaluates numerical radial derivatives on the solved samples,

$$C_i = g_i^2 + \kappa_{\text{struct}}^2 (D_r g)_i^2, \qquad u_i = |(D_r C)_i|. \tag{34}$$

Only samples with $r_i > R/5$ enter the outer structural readout. The registered derivative operator D_r is evaluated on the radius-sorted samples with the default numerical-gradient convention: centered finite differences in the interior and first-order one-sided differences at the two profile endpoints. If that outer window contains more than five samples, the registered weight is the equal-weight centered five-point moving average $w_i = (\text{MA}_5\, u)_i$, implemented as a same-length convolution; values beyond the two ends of the outer window are taken as zero by that convolution. Otherwise $w_i = u_i$. The sampled characteristic length and Structural Mode estimator are

$$\lambda_{\text{num}} = \frac{\sum_i r_i w_i}{\sum_i w_i}, \qquad N_{\text{num}} = \frac{R}{\lambda_{\text{num}}}. \tag{35}$$

The sums in Eq. (35) are the registered sampled-data estimator and do not introduce a Δr_i quadrature factor. The numerical Structural Mode values reported for DDO-154, NGC-3198, and NGC-7814 below are N_{num} outputs from this registered SPARC implementation.

The Curvature Structure Mode Index (CSMI) is a downstream descriptive catalog layer applied only after N_{num} has been calculated. Its named bands use the registered Workbench intervals given in the parent CPTG framework [9]; conventional visual morphology is not an input to the rotation-curve solve or to the Structural Mode calculation.

8 Example I: DDO-154 as a Low-Acceleration Dwarf

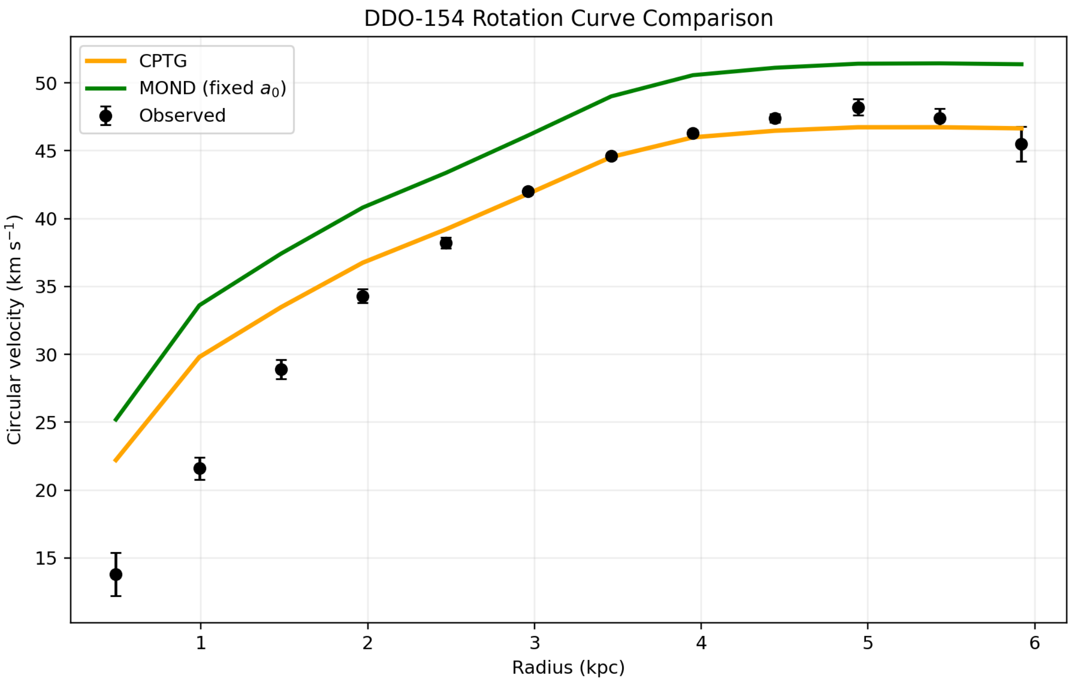

Figure 1: Rotation-curve comparison for DDO-154. Black markers and error bars show the observed rotation curve. The solid orange curve is the CPTG calculation, and the solid green curve is the fixed-a_0, unit-source MOND/RAR-form calculation defined in Eq. (18).

Figure 1 presents the full rotation-curve calculation before the numerical steps are unpacked. The remainder of the section follows the source data through the CPTG source construction, the inference of $a_\star$, the nonlinear field solve, and the final circular-velocity calculation.

8.1 Input Profile and Solved Scale

DDO-154 is a low-mass, gas-rich dwarf galaxy whose rotation curve has long been used as a sharp test of dark matter and modified-gravity models [3, 8]. Its stellar contribution is small, its gas contribution is dynamically important, and it sits deep in the low-acceleration regime. The SPARC mass-model entry for DDO-154 [5] gives a distance of 4.04 Mpc, with 12 rotation-curve points spanning 0.49 kpc to 5.92 kpc. The bulge component is zero throughout this SPARC rotation-curve decomposition:

$$V_{\rm bulge}(r) = 0. \tag{36}$$

The CPTG source field is therefore built from gas and disk only:

$$V^2_{\rm bar,CPTG}(r) = \frac{27}{50} V^2_{\rm gas}(r) + \frac{23}{50} V^2_{\rm disk}(r). \tag{37}$$

The inferred CPTG structural acceleration scale is

$$a_\star = 1.497936844 \times 10^{-10}\ \mathrm{m\ s^{-2}}. \tag{38}$$

The derived structural transport scale is

$$R_c = 9.047717100 \times 10^{28}\ \mathrm{m}. \tag{39}$$

8.2 Following the Calculation Across Three Radii

Table 1 follows the calculation vertically at three representative radii. The first block contains the observed and SPARC source velocities, followed by the observed acceleration as a separate diagnostic row. The CPTG acceleration-construction block then begins with $g_{\rm bar,CPTG}$, keeping $g_{\rm obs}$ visually separate from the pointwise theoretical solve. The final block returns the solved fields to circular velocity and places the fixed-a_0, unit-source comparison beside the CPTG result for the same radii.

Quantity	0.49 kpc	3.46 kpc	5.92 kpc
Observed and source velocities			$\mathrm{km\,s^{-1}}$
$V_{\rm obs}$	13.80	44.60	45.50
σ_V	1.60	0.20	1.30
$V_{\rm gas}$	3.74	17.60	15.64
$V_{\rm disk}$	12.31	8.37	6.26
$V_{\rm bulge}$	0.00	0.00	0.00
Observed acceleration			$\mathrm{m\,s^{-2}}$
$g_{\rm obs}$	1.259539×10^{-11}	1.863130×10^{-11}	1.133315×10^{-11}
CPTG acceleration construction			$\mathrm{m\,s^{-2}}$, except Σ
$g_{\rm bar,CPTG}$	5.109843×10^{-12}	1.868568×10^{-12}	8.217756×10^{-13}
$g_{\rm pol}$	2.766627×10^{-11}	1.673020×10^{-11}	1.109490×10^{-11}
$g_{\rm geom}$	1.537001×10^{-24}	2.371954×10^{-23}	5.030950×10^{-23}
Σ	0.994893	0.998709	0.999566
Δg	2.752497×10^{-11}	1.670861×10^{-11}	1.109009×10^{-11}
$g_{\rm CPTG}$	3.263481×10^{-11}	1.857717×10^{-11}	1.191186×10^{-11}
Calculated rotation curves			$\mathrm{km\,s^{-1}}$
$V_{\rm CPTG}$	22.21	44.54	46.65
$V_{\rm MOND}$	25.20	49.00	51.37

Table 1: Representative DDO-154 calculation traced through three radii. Reading downward follows the observational input, the separate observed-acceleration diagnostic, CPTG source and response construction, and the final velocity calculation. The MOND row gives the fixed-a_0, unit-source comparison calculated from the same SPARC component inputs using the MOND prescription defined earlier in the paper.

8.3 Reading a Representative Radius

At the outer listed radius, $r = 5.92$ kpc, the CPTG baryonic source field is

$$g_{\rm bar,CPTG} = 8.217756 \times 10^{-13} \text{ m s}^{-2}, \tag{40}$$

and the solved field is

$$g_{\rm CPTG} = 1.191186 \times 10^{-11} \text{ m s}^{-2}. \tag{41}$$

The additional response is supplied primarily by the nonlinear polarization term. At the same radius, the observed velocity is

$$V_{\rm obs} = 45.50 \text{ km s}^{-1}, \tag{42}$$

and the calculated CPTG velocity is

$$V_{\rm CPTG} = 46.65 \text{ km s}^{-1}. \tag{43}$$

8.4 Numerical Summary

Across the 12 measured radii, the CPTG calculation gives

$$\chi^2_{\mathrm{CPTG}} = 226.615133, \qquad \frac{\chi^2_{\mathrm{CPTG}}}{N_{\mathrm{data}}} = 18.884594, \qquad \chi^2_{\nu,\mathrm{CPTG}} = 20.601376.$$

Its velocity residuals have an RMS of 3.774060 km s^{-1} and a mean absolute error of 2.453007 km s^{-1}. The fixed-a_0, unit-source MOND/RAR-form calculation gives

$$\chi^2_{\mathrm{MOND}} = 2359.131043, \qquad \frac{\chi^2_{\mathrm{MOND}}}{N_{\mathrm{data}}} = 196.594254,$$

with an RMS residual of 6.740476 km s^{-1} and a mean absolute error of 6.101620 km s^{-1}.

In direct fractional-velocity terms, $\epsilon_{V,\mathrm{med}} = 2.588\%$ for CPTG and 11.386% for the comparison calculation. Nine of the 12 CPTG points and six of the 12 comparison points lie within 10% of the observed circular velocity.

The registered sampled-data structural readout is

$$N_{\mathrm{num}} = 1.890702489,$$

which lies in the registered CSMI *LSB Dwarf Disk* band.

DDO-154 illustrates the low-acceleration end of the calculation. Its weak baryonic source makes the square-root polarization contribution easy to trace directly from the source field to the solved velocity. The instructional sequence remains explicit: build the locked CPTG source, infer the galaxy's $a_\star$, derive R_c, solve the implicit field equation, convert the solved acceleration back to circular velocity, and then evaluate the post-solution Structural Mode.

9 Example II: NGC-3198 as an Extended Flat-Curve Spiral

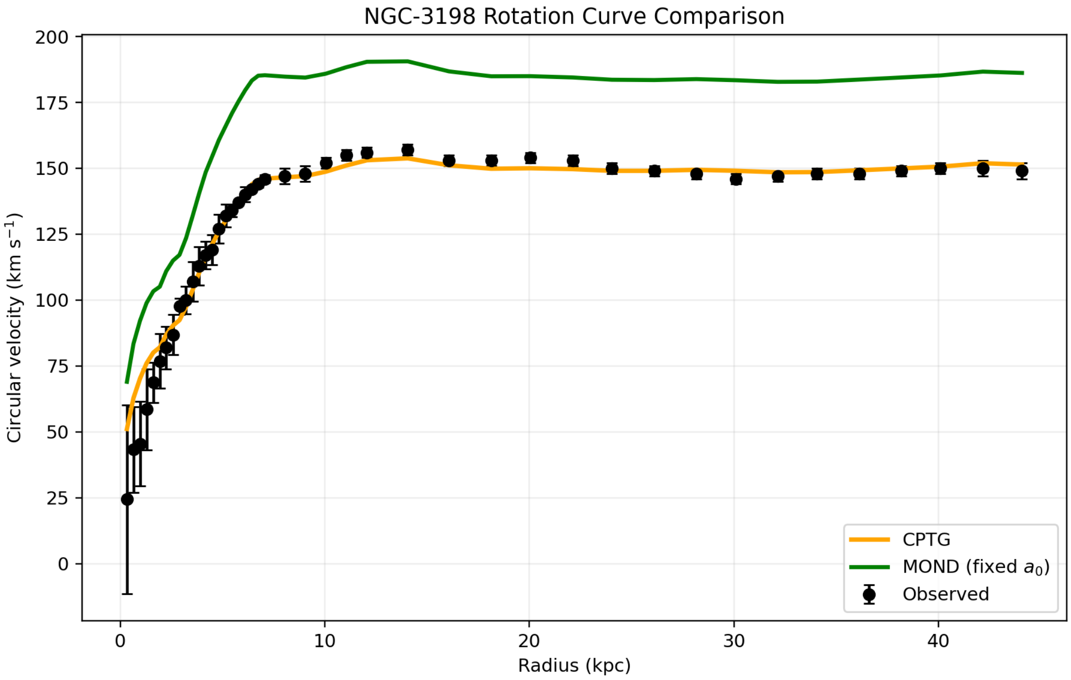

Figure 2: Rotation-curve comparison for NGC-3198. Black markers and error bars show the observed rotation curve. The solid orange curve is the CPTG calculation, and the solid green curve is the fixed-a_0, unit-source MOND/RAR-form calculation defined in Eq. (18).

Figure 2 presents the full rotation-curve calculation before the numerical steps are unpacked. The remainder of the section follows the same order used for DDO-154 so that the change from a dwarf source to an extended stellar disk can be read directly from the calculation.

9.1 Input Profile and Solved Scale

NGC-3198 is one of the classic extended rotation-curve galaxies, and its H I curve has long served as a standard reference for both dark-halo modeling and modified-gravity studies [1, 2]. The SPARC mass-model entry for NGC-3198 [5] gives a distance of 13.8 Mpc, with 43 rotation-curve points spanning 0.32 kpc to 44.08 kpc. The bulge component is zero throughout this SPARC rotation-curve decomposition:

$$V_{\text{bulge}}(r) = 0. \tag{44}$$

The CPTG source field is therefore built from gas and disk contributions:

$$V^2_{\text{bar,CPTG}}(r) = \frac{27}{50} V^2_{\text{gas}}(r) + \frac{23}{50} V^2_{\text{disk}}(r). \tag{45}$$

The inferred CPTG structural acceleration scale is

$$a_\star = 9.858387217 \times 10^{-11} \text{ m s}^{-2}. \tag{46}$$

The derived structural transport scale is

$$R_c = 1.374759228 \times 10^{29} \text{ m.} \tag{47}$$

9.2 Following the Calculation Across Three Radii

Table 2 uses the same vertical calculation order as Table 1. The inner column shows the stronger disk-source regime; the middle and outer columns show how the baryonic source weakens while the solved CPTG field maintains the extended rotation curve.

Quantity	**0.32 kpc**	**7.06 kpc**	**44.08 kpc**
Observed and source velocities			$\mathrm{km\,s^{-1}}$
V_{obs}	24.40	146.00	149.00
σ_V	35.90	1.57	3.00
V_{gas}	0.00	12.87	47.84
V_{disk}	63.28	140.94	61.63
V_{bulge}	0.00	0.00	0.00
Observed acceleration			$\mathrm{m\,s^{-2}}$
g_{obs}	6.029470×10^{-11}	9.784766×10^{-11}	1.632226×10^{-11}
CPTG acceleration construction			$\mathrm{m\,s^{-2}}$, except Σ
$g_{\mathrm{bar,CPTG}}$	1.865479×10^{-10}	4.235470×10^{-11}	2.193171×10^{-12}
g_{pol}	1.356120×10^{-10}	6.461803×10^{-11}	1.470412×10^{-11}
g_{geom}	3.101411×10^{-25}	2.358717×10^{-23}	3.063992×10^{-22}
Σ	0.575306	0.860100	0.997168
Δg	7.801835×10^{-11}	5.557799×10^{-11}	1.466249×10^{-11}
g_{CPTG}	2.645662×10^{-10}	9.793269×10^{-11}	1.685566×10^{-11}
Calculated rotation curves			$\mathrm{km\,s^{-1}}$
V_{CPTG}	51.11	146.06	151.42
V_{MOND}	69.01	185.31	186.18

Table 2: Representative NGC-3198 calculation traced through three radii. The table uses the same ordering as the DDO-154 example: observed and source velocities, the separate observed-acceleration diagnostic, CPTG acceleration construction, and the final CPTG and fixed-a_0, unit-source comparison values.

9.3 Reading a Representative Radius

At the outer listed radius, $r = 44.08$ kpc, the CPTG baryonic source field is

$$g_{\mathrm{bar,CPTG}} = 2.193171 \times 10^{-12} \text{ m s}^{-2}, \tag{48}$$

and the solved field is

$$g_{\mathrm{CPTG}} = 1.685566 \times 10^{-11} \text{ m s}^{-2}. \tag{49}$$

At the same radius, the observed velocity is

$$V_{\mathrm{obs}} = 149.00 \text{ km s}^{-1}, \tag{50}$$

and the calculated CPTG velocity is

$$V_{\rm CPTG} = 151.42~\mathrm{km~s^{-1}}. \tag{51}$$

9.4 Numerical Summary

Across the 43 measured radii, the CPTG calculation gives

$$\chi^2_{\rm CPTG} = 46.789053, \qquad \frac{\chi^2_{\rm CPTG}}{N_{\rm data}} = 1.088118, \qquad \chi^2_{\nu,\rm CPTG} = 1.114025.$$

Its velocity residuals have an RMS of 7.449345 km s^{-1} and a mean absolute error of 4.171527 km s^{-1}. The fixed-a_0, unit-source MOND/RAR-form calculation gives

$$\chi^2_{\rm MOND} = 12186.634554, \qquad \frac{\chi^2_{\rm MOND}}{N_{\rm data}} = 283.410106,$$

with an RMS residual of 35.003291 km s^{-1} and a mean absolute error of 34.617001 km s^{-1}.

In direct fractional-velocity terms, $\epsilon_{V,\rm med} = 1.621\%$ for CPTG and 24.602% for the comparison calculation. Thirty-eight of the 43 CPTG points and none of the 43 comparison points lie within 10% of the observed circular velocity.

The registered sampled-data structural readout is

$$N_{\rm num} = 3.062572658,$$

which lies in the registered CSMI *Early Spiral* band.

NGC-3198 demonstrates the large-disk version of the same calculation. The inner disk produces a stronger baryonic source field than DDO-154, so the nonlinear suppression factor is more active in the inner galaxy. Farther out, the baryonic source weakens, the suppression relaxes, and the square-root polarization response supplies the extended acceleration needed for the nearly flat measured curve.

10 Example III: NGC-7814 as a Bulge-Dominated Case

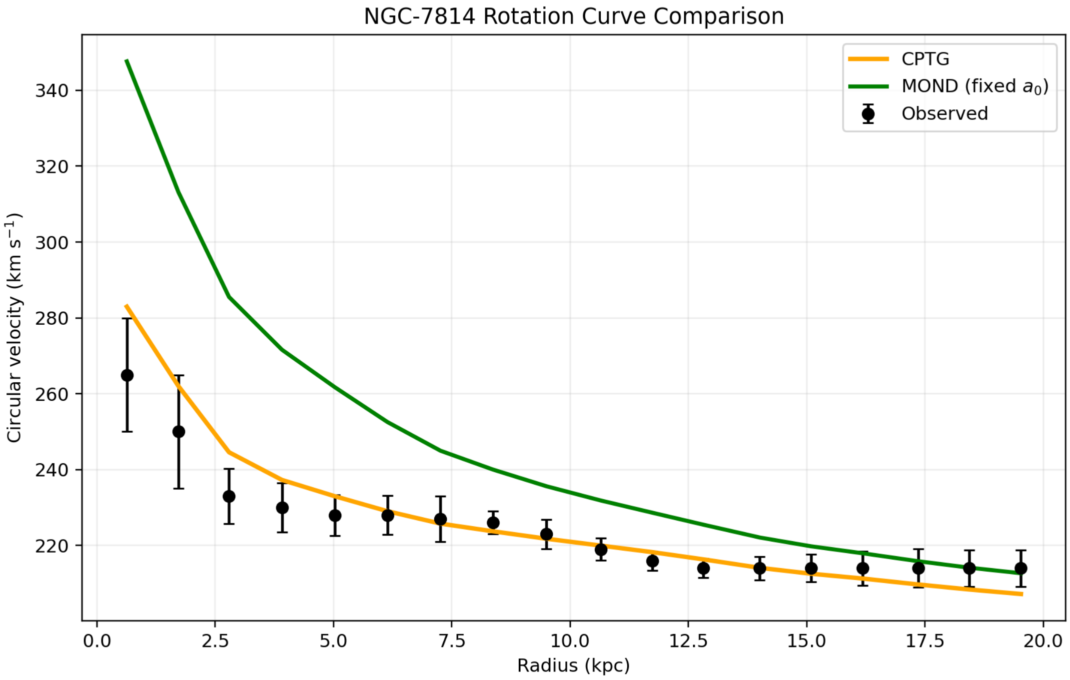

Figure 3: Rotation-curve comparison for NGC-7814. Black markers and error bars show the observed rotation curve. The solid orange curve is the CPTG calculation, and the solid green curve is the fixed-a_0, unit-source MOND/RAR-form calculation defined in Eq. (18).

Figure 3 presents the full rotation-curve calculation first. The section then follows the same mathematical order used for DDO-154 and NGC-3198, with the only source-level difference being the large nonzero bulge contribution.

10.1 Input Profile and Solved Scale

NGC-7814 is a centrally concentrated, bulge-dominated spiral whose light and mass decomposition make it a useful bulge-dominated example in alternative-gravity discussions [4]. The SPARC mass-model entry for NGC-7814 [5] gives a distance of 14.4 Mpc, with 18 rotation-curve points spanning 0.63 kpc to 19.53 kpc. Unlike DDO-154 and NGC-3198, its bulge component is nonzero throughout the measured radial range and dominates the inner decomposition: $V_{\text{bulge}} = 345.22$ km s^{-1} at the first listed radius and remains 105.49 km s^{-1} at the outer listed radius.

The CPTG source field therefore uses all three locked source components:

$$V^2_{\text{bar,CPTG}}(r) = \frac{27}{50}Q(V_{\text{gas}}(r)) + \frac{23}{50}Q(V_{\text{disk}}(r)) + \frac{16}{25}Q(V_{\text{bulge}}(r)) \,. \tag{52}$$

The inner gas entries in the SPARC decomposition are signed. The signed-square convention preserves that information inside the weighted source sum, so the negative inner gas entry subtracts from the intermediate source budget while the much larger positive disk and bulge terms keep the final scalar $g_{\mathrm{bar,CPTG}}$ positive.

The inferred CPTG structural acceleration scale is

$$a_\star = 1.830618925 \times 10^{-10} \text{ m s}^{-2}. \tag{53}$$

The derived structural transport scale is

$$R_c = 7.403457168 \times 10^{28} \text{ m}. \tag{54}$$

10.2 Following the Calculation Across Three Radii

Table 3 retains exactly the same row order as the previous two examples. This makes the effect of the bulge source and the stronger inner-field suppression directly comparable without changing the calculation sequence.

Quantity	**0.63 kpc**	**10.65 kpc**	**19.53 kpc**
Observed and source velocities			$\mathrm{km\,s^{-1}}$
V_{obs}	265.00	219.00	214.00
σ_V	15.00	2.90	4.87
V_{gas}	-0.69	16.09	22.64
V_{disk}	39.69	105.99	77.35
V_{bulge}	345.22	142.79	105.49
Observed acceleration			$\mathrm{m\,s^{-2}}$
g_{obs}	3.612440×10^{-9}	1.459446×10^{-10}	7.599320×10^{-11}
CPTG acceleration construction			$\mathrm{m\,s^{-2}}$, except Σ
$g_{\mathrm{bar,CPTG}}$	3.960827×10^{-9}	5.585805×10^{-11}	1.684440×10^{-11}
g_{pol}	8.515142×10^{-10}	1.011211×10^{-10}	5.552988×10^{-11}
g_{geom}	3.536121×10^{-24}	1.852421×10^{-22}	4.329457×10^{-22}
Σ	0.185423	0.903612	0.979619
Δg	1.578904×10^{-10}	9.137428×10^{-11}	5.439813×10^{-11}
g_{CPTG}	4.118717×10^{-9}	1.472323×10^{-10}	7.124253×10^{-11}
Calculated rotation curves			$\mathrm{km\,s^{-1}}$
V_{CPTG}	282.96	219.96	207.20
V_{MOND}	347.62	231.85	212.68

Table 3: Representative NGC-7814 calculation traced through three radii. The same table order is retained even though this galaxy contains a large bulge contribution and a signed inner gas term. The final row gives the fixed-a_0, unit-source comparison from the same SPARC component inputs.

10.3 Reading a Representative Radius

At the representative middle radius, $r = 10.65$ kpc, the CPTG baryonic source field is

$$g_{\mathrm{bar,CPTG}} = 5.585805 \times 10^{-11} \text{ m s}^{-2}, \tag{55}$$

and the solved field is

$$g_{\mathrm{CPTG}} = 1.472323 \times 10^{-10}\ \mathrm{m\ s^{-2}}. \tag{56}$$

At the same radius, the observed velocity is

$$V_{\mathrm{obs}} = 219.00\ \mathrm{km\ s^{-1}}, \tag{57}$$

and the calculated CPTG velocity is

$$V_{\mathrm{CPTG}} = 219.96\ \mathrm{km\ s^{-1}}. \tag{58}$$

The first table column shows the complementary high-field behavior. At $r = 0.63$ kpc, the large bulge term gives $g_{\mathrm{bar,CPTG}} = 3.960827 \times 10^{-9}$ m s^{-2} and the nonlinear suppression factor falls to $\Sigma = 0.185423$, limiting the polarization enhancement in the already strong inner field.

10.4 Numerical Summary

Across the 18 measured radii, the CPTG calculation gives

$$\chi^2_{\mathrm{CPTG}} = 13.762524, \qquad \frac{\chi^2_{\mathrm{CPTG}}}{N_{\mathrm{data}}} = 0.764585, \qquad \chi^2_{\nu,\mathrm{CPTG}} = 0.809560.$$

Its velocity residuals have an RMS of 6.687395 km s^{-1} and a mean absolute error of 4.780270 km s^{-1}. The fixed-a_0, unit-source MOND/RAR-form calculation gives

$$\chi^2_{\mathrm{MOND}} = 319.814608, \qquad \frac{\chi^2_{\mathrm{MOND}}}{N_{\mathrm{data}}} = 17.767478,$$

with an RMS residual of 31.882528 km s^{-1} and a mean absolute error of 22.268552 km s^{-1}.

In direct fractional-velocity terms, $\epsilon_{V,\mathrm{med}} = 1.178\%$ for CPTG and 5.859% for the comparison calculation. All 18 CPTG points and 12 of the 18 comparison points lie within 10% of the observed circular velocity.

The registered sampled-data structural readout is

$$N_{\mathrm{num}} = 3.157080935,$$

which lies in the registered CSMI *Early Spiral* band.

NGC-7814 demonstrates the bulge-dominated version of the same calculation. In the inner galaxy, the large bulge term produces a strong CPTG baryonic source field and the nonlinear suppression factor limits the polarization response. In the outer galaxy, the source field weakens and the polarization contribution becomes more important. The source composition changes, but the mathematical sequence does not.

11 Cross-Case Summary

The three worked examples contain 73 total rotation-curve points and span three different source regimes: a gas-rich dwarf, an extended stellar disk, and a bulge-dominated spiral. Table 4 collects the principal quantities in the same order used throughout the worked examples so that the reader can compare the calculations vertically rather than moving between separate summary tables.

Quantity	**DDO-154**	**NGC-3198**	**NGC-7814**
Observational scope and representative result			
Rotation-curve points	12	43	18
Representative radius (kpc)	5.92	44.08	10.65
$V_{\rm obs}$ (km s^{-1})	45.50	149.00	219.00
$V_{\rm CPTG}$ (km s^{-1})	46.65	151.42	219.96
CPTG structural scales			
$a_\star$ (m s^{-2})	$1.497936844 \times 10^{-10}$	$9.858387217 \times 10^{-11}$	$1.830618925 \times 10^{-10}$
R_c (m)	$9.047717100 \times 10^{28}$	$1.374759228 \times 10^{29}$	$7.403457168 \times 10^{28}$
CPTG numerical diagnostics			
$\chi^2_{\rm CPTG}$	226.615133	46.789053	13.762524
$\chi^2_{\rm CPTG}/N_{\rm data}$	18.884594	1.088118	0.764585
Reduced $\chi^2_{\rm CPTG}$	20.601376	1.114025	0.809560
CPTG RMS residual (km s^{-1})	3.774060	7.449345	6.687395
CPTG mean absolute error (km s^{-1})	2.453007	4.171527	4.780270
CPTG median absolute fractional velocity residual (%)	2.588	1.621	1.178
CPTG points within 10% of $V_{\rm obs}$	9/12	38/43	18/18
Fixed-a_0, unit-source MOND/RAR-form comparison			
$\chi^2_{\rm MOND}$	2359.131043	12186.634554	319.814608
$\chi^2_{\rm MOND}/N_{\rm data}$	196.594254	283.410106	17.767478
MOND RMS residual (km s^{-1})	6.740476	35.003291	31.882528
MOND mean absolute error (km s^{-1})	6.101620	34.617001	22.268552
MOND median absolute fractional velocity residual (%)	11.386	24.602	5.859
MOND points within 10% of $V_{\rm obs}$	6/12	0/43	12/18
Post-solution structural readout			
Workbench Structural Mode $N_{\rm num}$	1.890702489	3.062572658	3.157080935
CSMI	LSB Dwarf Disk	Early Spiral	Early Spiral

Table 4: Cross-case calculation summary. Reading downward follows the same sequence used in the three worked examples: observational scope and representative velocity result, CPTG structural scales, CPTG numerical diagnostics including direct fractional-velocity agreement, the fixed-a_0, unit-source MOND/RAR-form comparison, and the post-solution sampled Structural Mode/CSMI readout. The CPTG reduced χ^2 values use one inferred parameter, $a_\star$; the MOND calculation retains the fixed universal a_0 defined in Eq. (18).

Across all 73 measured rotation-curve points, the median absolute fractional velocity residual is 1.897% for CPTG and 23.479% for the fixed-a_0, unit-source MOND/RAR-form comparison. Sixty-five of 73 CPTG points (89.0%) and 18 of 73 comparison points (24.7%)

lie within 10% of the observed circular velocity. These pointwise fractions provide a direct descriptive measure of how closely each calculated curve follows the measured rotation speed, complementary to the uncertainty-weighted χ^2 statistics.

The comparison makes the common calculation architecture explicit. DDO-154 shows the weak-source dwarf calculation, NGC-3198 the extended-disk calculation, and NGC-7814 the bulge-dominated calculation. In every case the source coefficients, transport coefficient, transport relation, and nonlinear exponents remain locked. The changing inputs are the observed source decomposition and the object-dependent $a_\star$ inferred for that galaxy.

All measured radii in these three examples lie far inside their derived R_c values. Consequently, the worked calculations primarily illustrate the locked source mapping, square-root polarization response, and nonlinear suppression; the direct geometric transport remnant $g_{\rm geom}$ remains a small contribution throughout the measured radial ranges.

12 Galaxy Analysis with the CPTG SPARC Browser Workbench

The CPTG SPARC Browser Workbench provides a public, local interface for applying the galaxy-scale calculation described in this tutorial across the SPARC rotation-curve database. It is intended to let a reader move from the three worked examples in this paper to direct single-galaxy or multi-galaxy exploration without having to reconstruct the calculation pipeline manually.

The standalone workbench includes the SPARC galaxy files and supporting metadata needed to begin analysis. It provides searchable galaxy selection, individual and batch calculations, metadata views for primary, excluded, and unmatched systems, CPTG and MOND rotation-curve comparisons, averaged normalized rotation curves, averaged radial-acceleration-relation scatter, compact fit summaries, and galaxy-level Structural Mode and CSMI results. Optional CSV, JSON, PNG, and ZIP outputs can be saved when a persistent record is desired. Calculations are performed locally, and completed runs are displayed directly in the interface unless the user explicitly chooses to save result files. The workbench is maintained with the public CPTG repository [10].

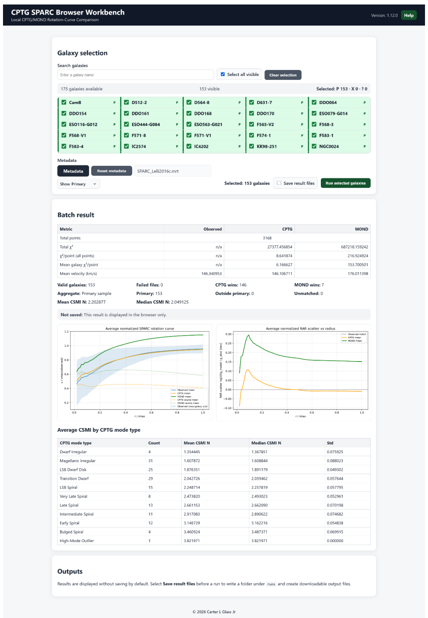

Figure 4: CPTG SPARC Browser Workbench in batch-analysis mode. Interface shown for illustration; the current release may differ.

References

[1] K. G. Begeman, "H I rotation curves of spiral galaxies. I. NGC 3198," *Astronomy and Astrophysics*, **223**, 47–60 (1989).

[2] K. G. Begeman, A. H. Broeils, and R. H. Sanders, "Extended rotation curves of spiral galaxies: dark haloes and modified dynamics," *Monthly Notices of the Royal Astronomical Society*, **249**, 523–537 (1991).

[3] C. Carignan, "DDO 154: A dark galaxy?," *The Astrophysical Journal Letters*, **332**, L33–L36 (1988).

[4] F. Fraternali, R. Sancisi, and P. Kamphuis, "A tale of two galaxies: light and mass in NGC 891 and NGC 7814," *Astronomy and Astrophysics*, **531**, A64 (2011).

[5] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *The Astronomical Journal*, **152**, 157 (2016).

[6] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters*, **117**, 201101 (2016).

[7] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," *The Astrophysical Journal*, **270**, 365–370 (1983).

[8] M. Milgrom, "The rotation curve of DDO 154: A particularly acute test of the modified dynamics," *The Astrophysical Journal*, **334**, 130–134 (1988).

[9] Carter L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, 2026, DOI: 10.5281/zenodo.22441039.

[10] *CPTG, Supporting Python Models, Galaxy-Calculation Implementations, and Research References for Curvature Polarization Transport Gravity*, companion resource for the present work.